Math

Q.1

Ans 3 Zeroes

Q.2) $3x^2 - Kx + 6$

Ans

Sum of Zeroes $= \frac{-b}{a}$

$a = 3$, $b = -K$ $c = 6$

Sum $= \frac{-(-K)}{3} = \frac{K}{3}$

$\frac{K}{3} \times \frac{3}{1} = K = 9$

Q.3)

Ans Trivariate

Q.4)

$\rightarrow$

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Q.5)

→ $\alpha+\beta = \frac{-1}{2}$, $\alpha\beta = -3$

$x^2-(\alpha+\beta)x+\alpha\beta$

$x^2-\left(\frac{-1}{2}\right)x+-3$

$\Rightarrow 2x^2-x-6$

Q.6)

→ $a=1$, $b=-5$, $c=6$

$\alpha+\beta = \frac{-b}{a} = \frac{-5}{1} = 5$

$\alpha\beta = \frac{c}{a} = \frac{6}{1} = 6$

$(\alpha+\beta)^2 = \alpha^2+\beta^2+2ab$

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$(5)^2 = a^2 + B^2 + 2(6)$

$25 = a^2 + B^2 + 12$

$25 - 12 = a^2 + \beta^2$

$13 = a^2 + b^2$

$\alpha^2 + \beta^2 = 13$

Q.7)

$\rightarrow x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$\alpha + \beta = \frac{-4}{4} = -1$

$\alpha\beta = \frac{1}{4}$

$\Rightarrow$ Zeroes are 2α and 2β

$\therefore 2\alpha + 2\beta = 2(\alpha + \beta) = 2(-1) = -2$

$\therefore 2\alpha \times 2\beta = 2\alpha\beta = 4 \times \left[\frac{1}{4}\right] = 1$

$x^2 - (-2)x - 1 = 0$

$x^2 + 2x + 1 = 0$

Q8)

$\rightarrow$ $x = 1$ --- (given)

$p(1) = a(1)^2 - 3(a-1)(1) - 1 = 0$

$= a - 3(a-1) - 1 = 0$

$a - 3a + 3 - 1 = 0$

$-2a + 2 = 0$

$\rightarrow$ $-2a = -2$

$a = \frac{\cancel{-2}}{\cancel{-2}}$

$a = 1$

Q.9) $4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 0$

5

3 2

$4x(\sqrt{3}x + 2) - \sqrt{3}(\sqrt{3}x + 2) = 0$

$4x - \sqrt{3})(\sqrt{3}x + 2) = 0$

$\sqrt{3}x + 2 = 0$

$x = \frac{-2}{\sqrt{2}}$

$4x - \sqrt{3} = 0$

$x = \frac{\sqrt{3}}{4}$

Q.10)

→ $\alpha + \beta = 6$ -- (i)

$\alpha\beta = K$ -- (ii)

$3\alpha + 2\beta = 20$ -- (given)

$3(6-\beta) + (2\beta) = 20$

$18 - 3\beta + 2\beta = 20$

$-\beta = 2$

$\beta = -2$

from (i)

$\alpha = 8$

from (ii)

$\alpha\beta = K$

$(8)(-2) = K$

$K = -16$